

Definition 1. Let Var be the set of propositional variables. Then ϕ is defined inductively as follows:

$$\phi ::= p \mid \phi \vee \phi \mid X\phi \mid \phi U \psi \mid \langle a \rangle \phi$$

Definition 2. Let ϕ be a formula and $p \in Var$. Then the Closure of ϕ is inductively defined as

$$\begin{aligned} Cl(p) &= Cl(\neg p) = \{p, \neg p\} \\ Cl(\neg \phi) &= Cl(\phi) \\ Cl(\phi \vee \psi) &= \{\phi \vee \psi, \neg(\phi \vee \psi)\} \cup Cl(\phi) \cup Cl(\psi) \\ Cl(X\phi) &= \{X\phi, \neg X\phi\} \cup Cl(\phi) \\ Cl(\phi U \psi) &= \{\phi U \psi, \neg(\phi U \psi)\} \cup Cl(\phi) \cup Cl(\psi) \\ Cl(\langle a \rangle \phi) &= \{\langle a \rangle \phi, \neg \langle a \rangle \phi\} \cup Cl(\phi) \end{aligned}$$

Definition 3. An atom A is a subset of $Cl(\phi)$ such that

$$\begin{aligned} \text{if } \psi \in Cl(\phi), & \quad \text{then } \psi \in A \text{ iff } \neg\psi \notin A, \\ \text{if } \phi_1 \vee \phi_2 \in Cl(\phi), & \quad \text{then } \phi_1 \vee \phi_2 \in A \text{ iff } \phi_1 \in A \text{ or } \phi_2 \in A \\ \text{if } \phi_1 U \phi_2 \in Cl(\phi), & \quad \text{then if } \phi_1 U \phi_2 \in A \text{ then } \psi_1 \in A \text{ or } \psi_2 \in A \\ \text{if } \phi_1 U \phi_2 \in Cl(\phi), & \quad \text{then if } \phi_2 \in A \text{ then } \phi_1 U \phi_2 \in A \end{aligned}$$

Definition 4. We say two atoms A, B are in relation $R_x(AR_x B)$ iff

$$\begin{aligned} \text{if } X\psi \in Cl(\phi), & \quad \text{then } X\psi \in A \text{ iff } \psi \in B, \\ \text{if } \psi_1 U \psi_2 \in Cl(\phi), & \quad \text{then if } \psi_1 U \psi_2, \neg\psi_2 \in A, \text{ then } \psi_1 U \psi_2 \in B, \\ & \quad \text{and if } \psi_1 U \psi_2 \in B, \psi_1 \in A, \text{ then } \psi_1 U \psi_2 \in A. \end{aligned}$$

Lemma 5. For any atom A there exists an atom B s.t. $AR_x B$.

Proof. Assume A is an atom. Then we define B' as follows:

$$B' = \{\phi \mid X\phi \in A\} \cup \{\neg\phi \mid \neg X\phi \in A\} \cup \{\psi_1 U \psi_2 \mid \psi_1 U \psi_2, \neg\psi_2 \in A\} \cup \{\neg(\psi_1 U \psi_2) \mid \neg(\psi_1 U \psi_2), \psi_1 \in A\}$$

Now we show that B is consistent. Assume it is not. Then there exists $\alpha_1, \dots, \alpha_n$ s.t. $B \vdash \alpha_1 \wedge \alpha_2 \dots \wedge \alpha_n \rightarrow \perp$. Let's define $\hat{B} = \bigwedge B$. It follows:

$$\begin{aligned} & \vdash \hat{B} \wedge \alpha_1 \wedge \dots \wedge \alpha_n \rightarrow \perp \\ & \vdash X(\hat{B} \wedge \alpha_1 \wedge \dots \wedge \alpha_n \rightarrow \perp) && \text{By X-necessitation} \\ & \vdash X(\hat{B} \wedge \alpha_1 \wedge \dots \wedge \alpha_n) \rightarrow X\perp && \text{F2} \\ & \vdash X(\hat{B}) \wedge X\alpha_1 \wedge \dots \wedge X\alpha_n \rightarrow X\perp \end{aligned}$$

Assume, $a_i = \phi$. Then by definition we have $A \vdash Xa_i$. If $a_i = \neg\phi$, then we can use F1 to obtain $A \vdash X\neg\phi$. If $a_i = \psi_1 U \psi_2$ then by definition, $\psi_1 U \psi_2 \in A$ and $\neg\psi_2 \in A$. Consider the following proof steps:

$$\begin{aligned}
& A \vdash \neg\psi_2 \\
& A \vdash \psi_1 U \psi_2 \\
& A \vdash \psi_1 U \psi_2 \wedge \neg\psi_2 \\
& A \vdash \psi_1 U \psi_2 \wedge \neg\psi_2 \rightarrow (\psi_2 \vee (\psi_1 \wedge X(\psi_1 U \psi_2))) \wedge \neg\psi_2 \quad \text{Unfolding U} \\
& A \vdash \psi_1 U \psi_2 \wedge \neg\psi_2 \rightarrow (\psi_1 \wedge X(\psi_1 U \psi_2)) \quad \text{distributivity and other rules} \\
& A \vdash \psi_1 \wedge X(\psi_1 U \psi_2) \\
& A \vdash X(\psi_1 U \psi_2).
\end{aligned}$$

Hence, $A \vdash Xa_i$. If $a_i = \neg(\psi_1 U \psi_2)$, then by definition, $\neg(\psi_1 U \psi_2), \psi_1 \in A$. Our goal is to derive $A \vdash \neg(\psi_1 U \psi_2) \wedge \psi_1 \rightarrow X\neg(\psi_1 U \psi_2)$.

$$\begin{aligned}
& A \vdash \neg(\psi_1 U \psi_2) \wedge \psi_1 \\
& A \vdash \neg(\psi_2 \vee (\psi_1 \wedge X(\psi_1 U \psi_2))) \wedge \psi_1 \quad \text{Unfolding U} \\
& A \vdash \neg\psi_2 \wedge (\neg\psi_1 \vee \neg X(\psi_1 U \psi_2)) \wedge \psi_1 \\
& A \vdash \neg\psi_1 \vee \neg X(\psi_1 U \psi_2) \\
& \neg X(\psi_1 U \psi_2) \quad \psi_1 \text{ is derivable} \\
& A \vdash X\neg(\psi_1 U \psi_2) \quad \text{F1}
\end{aligned}$$

By similar argument, we can conclude $A \vdash X\hat{B}$. So it follows

$$A \vdash X\perp$$

Because $X\perp \rightarrow \perp$ is a tautology, we conclude $A \vdash \perp$ which is a contradiction. Now, by the Extension Lemma, we can extend B' to an atom B . It is easy to check, that $AR_x B$.

□